

USER INTERFACE DESIGN

EXPERIMENT 3

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Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)

1.Command Line Interface

A Command Line Interface (CLI) allows the user to interact with the system by typing commands and inputs through the keyboard.

It is simple and fast because the user directly enters data without using buttons or graphics.

CLI programs require less memory and system resources compared to graphical applications.

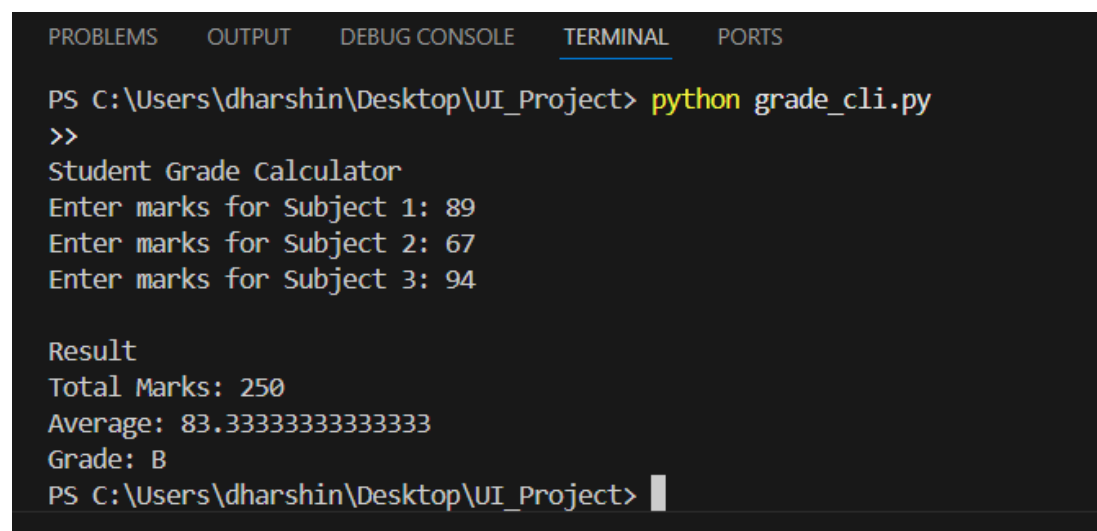
In this experiment, the user enters subject marks through the terminal, and the system calculates and displays the grade using text-based interaction.

IMPLEMENTATION

```
print("Student Grade Calculator")
m1 = int(input("Enter marks for Subject 1: "))
m2 = int(input("Enter marks for Subject 2: "))
m3 = int(input("Enter marks for Subject 3: "))
```

```
total = m1 + m2 + m3
average = total / 3
if average >= 90:
    grade = "A"
elif average >= 75:
    grade = "B"
elif average >= 50:
    grade = "C"
else:
    grade = "Fail"
print("\nResult")
print("Total Marks:", total)
print("Average:", average)
print("Grade:", grade)
```

OUTPUT:



The screenshot shows a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active and underlined), and 'PORTS'. The terminal content shows the command 'python grade_cli.py' being executed. The program prompts for marks for three subjects: 89, 67, and 94. It then displays the results: Total Marks: 250, Average: 83.33333333333333, and Grade: B. The prompt 'PS C:\Users\dhharshin\Desktop\UI_Project>' is visible at the bottom.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\dhharshin\Desktop\UI_Project> python grade_cli.py
>>
Student Grade Calculator
Enter marks for Subject 1: 89
Enter marks for Subject 2: 67
Enter marks for Subject 3: 94

Result
Total Marks: 250
Average: 83.33333333333333
Grade: B
PS C:\Users\dhharshin\Desktop\UI_Project> |
```

2.Graphical User Interface

A Graphical User Interface (GUI) allows users to interact with the system using visual elements like windows, buttons, and text boxes instead of typing commands.

It is user-friendly and easy to understand because users can click buttons and enter data in input fields.

GUI provides a better visual experience and is suitable for beginners who may not know command-line instructions.

In this experiment, the user enters subject marks in text boxes, clicks a button, and the system displays the total, average, and grade in a window.

IMPLENTATION:

```
import tkinter as tk  
from tkinter import messagebox  
  
def calculate():  
    try:  
        m1 = int(entry1.get())  
        m2 = int(entry2.get())  
        m3 = int(entry3.get())  
  
        total = m1 + m2 + m3  
        average = total / 3
```

```
if average >= 90:
    grade = "A"
elif average >= 75:
    grade = "B"
elif average >= 50:
    grade = "C"
else:
    grade = "Fail"

result_label.config(
    text=f"Total: {total} Average: {average:.2f} Grade:
{grade}"
)

except:
    messagebox.showerror("Error", "Please enter valid
numbers!")

# Window
root = tk.Tk()
root.title("Student Grade Calculator")
root.geometry("350x300")

tk.Label(root, text="Enter Subject 1 Marks").pack()
```

```
entry1 = tk.Entry(root)  
entry1.pack()
```

```
tk.Label(root, text="Enter Subject 2 Marks").pack()  
entry2 = tk.Entry(root)  
entry2.pack()
```

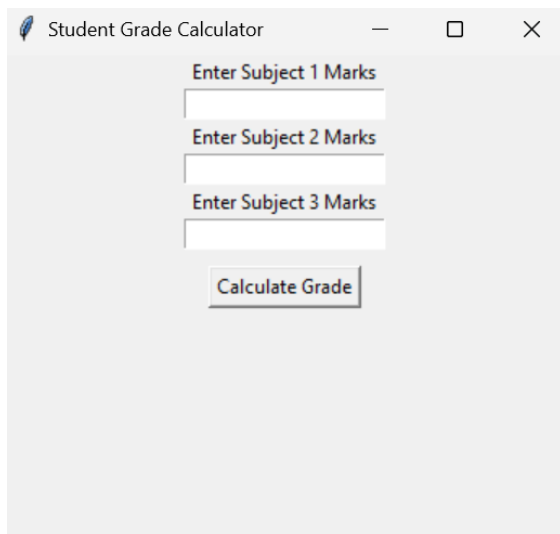
```
tk.Label(root, text="Enter Subject 3 Marks").pack()  
entry3 = tk.Entry(root)  
entry3.pack()
```

```
tk.Button(root, text="Calculate Grade",  
command=calculate).pack(pady=10)
```

```
result_label = tk.Label(root, text="")  
result_label.pack()
```

```
root.mainloop()
```

OUTPUT:



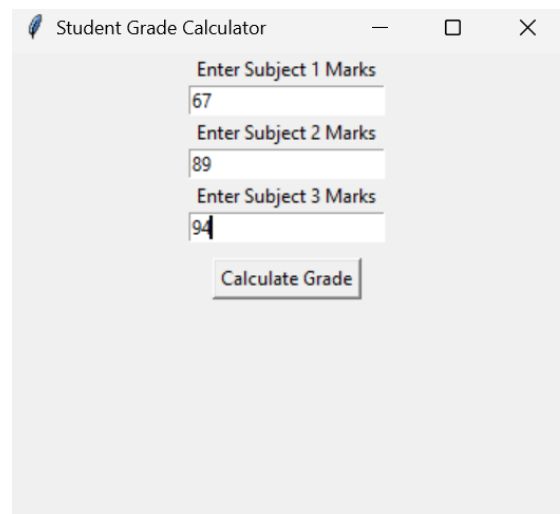
Student Grade Calculator

Enter Subject 1 Marks

Enter Subject 2 Marks

Enter Subject 3 Marks

Calculate Grade



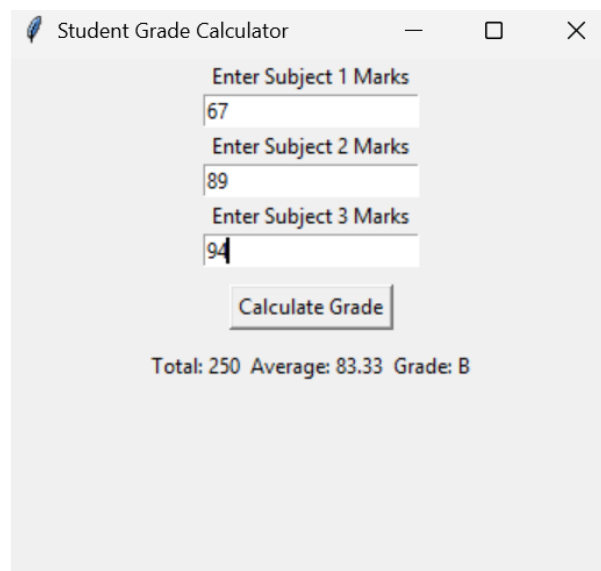
Student Grade Calculator

Enter Subject 1 Marks

Enter Subject 2 Marks

Enter Subject 3 Marks

Calculate Grade



Student Grade Calculator

Enter Subject 1 Marks

Enter Subject 2 Marks

Enter Subject 3 Marks

Calculate Grade

Total: 250 Average: 83.33 Grade: B

3.Voice User Interface

A Voice User Interface (VUI) allows users to interact with the system using voice commands instead of typing or clicking.

It makes interaction hands-free and more natural, as users can simply speak to give input.

VUI uses speech recognition technology to convert spoken words into text and process the request.

In this experiment, the user speaks the marks through a microphone, and the system recognizes the voice input, calculates the grade, and displays the result.

IMPLEMENTATION

```
import time

print("🎤 Voice User Interface Activated")

time.sleep(0.5)

print("& System: Welcome! I will calculate your grade for three subjects.")

time.sleep(0.5)

while True:

    print("@ System: Please say your marks for three subjects separated by space (e.g., 85 90 78)")

    marks_input = input("You: ")

    print("> Processing your voice input...")

    time.sleep(0.5)
```

```
try:
    marks = list(map(int, marks_input.split()))
    if len(marks) != 3:
        print("& System: Please say exactly three numbers.")
        continue
except ValueError:
    print("& System: Please say valid numbers only.")
    continue
average = sum(marks)/3
if average >= 90:
    grade = "A"
elif average >= 75:
    grade = "B"
elif average >= 60:
    grade = "C"
else:
    grade = "F"
print(f"& System: Your average is {average:.2f} and your grade is {grade}")
break
```

OUTPUT:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\dharshin\Desktop\UI_Project> python grade_vui.py
👉 Voice User Interface Activated
& System: Welcome! I will calculate your grade for three subjects.
@ System: Please say your marks for three subjects separated by space (e.g., 85 90 78)
You: 67 87 97
> Processing your voice input...
& System: Your average is 83.67 and your grade is B
PS C:\Users\dharshin\Desktop\UI_Project> |
```

4.User Satisfaction Comparison

```
import time
```

```
print("📊 User Satisfaction Comparison")
```

```
time.sleep(0.5)
```

```
print("Rate each interface from 1 (Very Poor) to 5 (Excellent)")
```

```
print("-" * 45)
```

```
time.sleep(0.5)
```

```
def get_rating(interface_name, emoji):
```

```
    while True:
```

```
        try:
```

```
            rating = int(input(f"{emoji} Rate {interface_name} (1-5):"))
```

```
            if 1 <= rating <= 5:
```

```
                return rating
```

```
            else:
```

```
print(f"& System: Please enter a number between 1 and 5 for {interface_name}.")
```

```
except ValueError:
```

```
print(f"& System: Invalid input. Enter a number between 1 and 5.")
```

```
cli = get_rating("CLI", "💻 ")
```

```
gui = get_rating("GUI", "🖥 ")
```

```
vui = get_rating("VUI", "🎤 ")
```

```
average = (cli + gui + vui) / 3
```

```
print("\n📊 Final Ratings Summary")
```

```
print("-" * 45)
```

```
print(f"💻 CLI Rating : {cli}/5")
```

```
print(f"🖥 GUI Rating : {gui}/5")
```

```
print(f"🎤 VUI Rating : {vui}/5")
```

```
print("-" * 45)
```

```
print(f"★ Average User Satisfaction : {average:.2f}/5")
```

```
print("\n🤖 Interpretation:")
```

```
if average >= 4:
```

```
    print("Users are highly satisfied with the interfaces 🙌")
```

```
elif average >= 3:
```

```
    print("Users are moderately satisfied 😊")
```

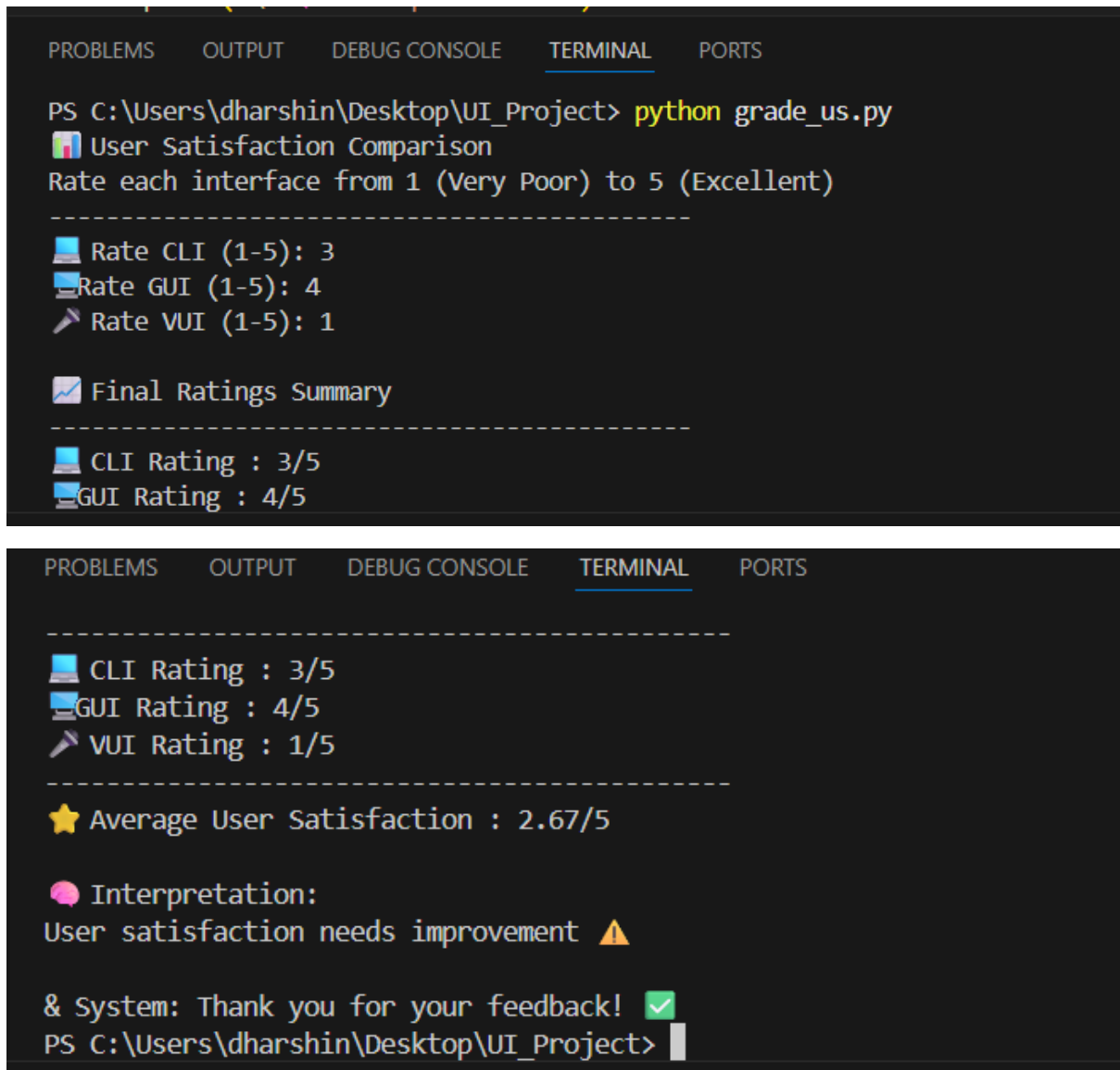
```
else:
```

```
    print("User satisfaction needs improvement ⚠")
```

```
time.sleep(0.5)
```

```
print("\n& System: Thank you for your feedback! ☑")
```

OUTPUT:



The image shows two screenshots of a terminal window displaying the output of a Python program. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The first screenshot shows the program's execution starting with the command `python grade_us.py`. The output includes a title "User Satisfaction Comparison", a prompt to rate interfaces, and three input prompts for CLI, GUI, and VUI ratings. The second screenshot shows the program's final output, including a "Final Ratings Summary" section with the ratings for CLI (3/5), GUI (4/5), and VUI (1/5). It also displays the "Average User Satisfaction" as 2.67/5 and an "Interpretation" that "User satisfaction needs improvement". The program concludes with a thank you message and a green checkmark icon.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\dhharshin\Desktop\UI_Project> python grade_us.py
📄 User Satisfaction Comparison
Rate each interface from 1 (Very Poor) to 5 (Excellent)
-----
📄 Rate CLI (1-5): 3
📄 Rate GUI (1-5): 4
🔧 Rate VUI (1-5): 1

📄 Final Ratings Summary
-----
📄 CLI Rating : 3/5
📄 GUI Rating : 4/5

-----
📄 CLI Rating : 3/5
📄 GUI Rating : 4/5
🔧 VUI Rating : 1/5
-----
★ Average User Satisfaction : 2.67/5

🧠 Interpretation:
User satisfaction needs improvement ⚠

& System: Thank you for your feedback! ☑
PS C:\Users\dhharshin\Desktop\UI_Project>
```